

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

MACHINE LEARNING-BASED STUDENT PERFORMANCE AND DROPOUT RISK PREDICTION FOR EARLY IDENTIFICATION OF AT-RISK STUDENTS

SDG 4 – Quality Education | SDG 10 – Reduced Inequalities

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INTRODUCTION

- Student academic performance is an important indicator of educational success.
 - Students may experience academic difficulties due to poor grades, low study time, attendance issues, previous failures and other social or academic factors.
 - Early identification of at-risk students helps institutions provide timely academic support and counselling.
 - Machine Learning can analyze historical student data and identify patterns associated with different performance levels.
 - This project develops an ML-based student performance prediction system that classifies students into performance categories.
 - **Importance:**
 - Identifies academically vulnerable students early.
 - Supports personalized academic intervention.
 - Helps institutions monitor performance continuously.
 - Reduces the chance of students drifting toward failure or dropout.
 - Provides a data-driven approach instead of manual observation alone.
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PROJECT OVERVIEW & OBJECTIVES / SCOPE

Project Overview

- ☐ **An integrated AI-based system for student performance and dropout-risk prediction with early risk identification, explainability and personalized intervention.**

- ☐ **Key Objectives**
 - ☐ • **Manage student data and preprocess datasets.**
 - ☐ • **Train and compare multiple ML models and select the best model.**
 - ☐ • **Predict High / Medium / Low academic performance.**
 - ☐ • **Predict Low / Medium / High dropout risk.**
 - ☐ • **Identify at-risk students using risk scores and trends.**
 - ☐ • **Explain predictions and provide personalized interventions.**

- ☐ **Scope**
 - ☐ **Included: student management, dataset preprocessing, model comparison, performance prediction, dropout prediction, risk analysis, XAI, interventions, reports and administration.**
 - ☐ **Implemented: early deterioration detection, hidden/emerging risk detection, risk trajectory analysis, SHAP-based explainability, personalized intervention and reports/administration.**

EXISTING SYSTEM

Existing Approaches

- Performance commonly monitored using exam marks and attendance.
- Faculty manually identify students needing additional support.
- Traditional systems mainly rely on historical academic records.
- Some existing ML systems predict performance or dropout risk using classification models.

Limitations

- Manual identification is time-consuming and observation-dependent.
- Multiple student-related factors are often not considered together.
- Some systems focus only on final grades; class imbalance hurts predictions.
- Few approaches compare multiple ML algorithms or apply hyperparameter tuning.
- Lack of explainability makes predictions hard for educators to interpret.

Proposed Improvement: combine academic, behavioural, demographic & social features; compare six ML models; select a tuned model.

PROBLEM STATEMENT

Problem

- ☐ Institutions often struggle to identify students with poor academic performance or increasing dropout risk at an early stage. Traditional monitoring may not capture combined academic patterns, emerging deterioration and changing risk over time.

 - ☐ Why It Matters
 - ☐ • Delayed intervention makes problems harder to correct.
 - ☐ • Students needing support may not be identified in time.
 - ☐ • Institutions need reliable, data-driven monitoring.

 - ☐ Proposed Solution
 - ☐ An integrated AI-based system that predicts performance and dropout risk, detects early deterioration and emerging risk, tracks risk trajectories, explains predictions and recommends personalized interventions.
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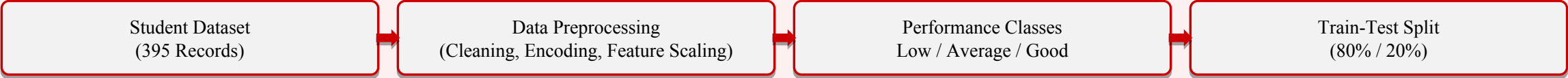
METHODOLOGY

1. Dataset Management – Upload and manage student datasets.

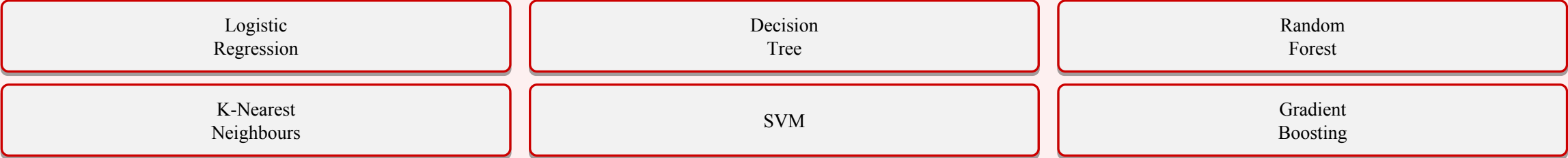
- ☐ **2. Data Preprocessing – Cleaning, missing values, duplicates, encoding and scaling.**
 - ☐ **3. ML Model Training – Train multiple models for performance and dropout prediction.**
 - ☐ **4. Model Evaluation & Comparison – Compare metrics and select the best model.**
 - ☐ **5. Performance Prediction – Predict High / Medium / Low performance.**
 - ☐ **6. Dropout Prediction – Predict Low / Medium / High dropout risk.**
 - ☐ **7. Risk Analysis – Generate risk scores, identify at-risk students and track trajectories.**
 - ☐ **8. Explainable AI – Use SHAP and feature importance to explain risk.**
 - ☐ **9. Personalized Intervention – Generate interventions based on actual risk factors.**
 - ☐ **10. Reports & Administration – Reports, user/model management, audit logs and settings.**
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SYSTEM ARCHITECTURE

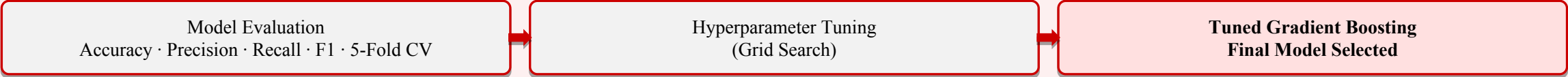
1 DATA PREPARATION



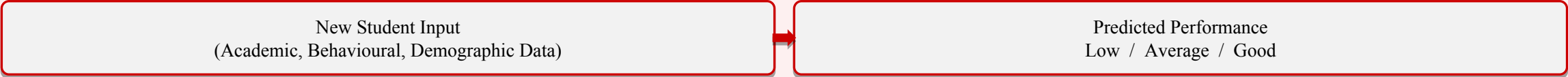
2 MODEL TRAINING — SIX ALGORITHMS COMPARED IN PARALLEL



3 EVALUATION & HYPERPARAMETER TUNING



4 PREDICTION ON NEW STUDENTS



TOOLS, TECHNOLOGIES AND DATASET

Programming Language: Python

Development Environment: Google Colab, Jupyter Notebook

Libraries: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn

ML Algorithms: Logistic Regression, Decision Tree, Random Forest, KNN, SVM, Gradient Boosting

Dataset: Student Performance Dataset — see summary table

Metric	Value
Total records	395
Original features	33
Input features (target removed)	32
Training records	316
Testing records	79
Processed features	58

END-TO-END USE CASE – LIVE DEMO

Use Case Step	Module(s)	Status	Notes
1. Load student dataset	Dataset Module	Live	395 records loaded
2. Data preprocessing	Preprocessing Module	Live	Categorical & numerical features processed
3. Performance classification	Classification Module	Live	Low, Average, Good classes
4. Train ML models	Model Training Module	Live	Six algorithms trained
5. Compare models	Evaluation Module	Live	Accuracy, Precision, Recall, F1
6. 5-Fold Cross Validation	Validation Module	Live	Model stability evaluated
7. Hyperparameter tuning	Optimization Module	Live	Grid Search performed
8. Select final model	Model Selection Module	Live	Tuned Gradient Boosting selected
9. Enter new student details	Prediction Module	Live	User provides student info
10. Predict performance	Prediction Module	Live	Predicts Low / Average / Good
11. Display probability	Prediction Module	Live	Per-class probability shown
12. Feature importance	Explainability Module	Live	Key predictive features identified

RESULTS & DISCUSSION

Initial 4-Class Model — relatively poor performance

Model	Accuracy	F1-Score
Logistic Regression	49.37%	48.94%
Gradient Boosting	46.84%	45.00%
Random Forest	46.84%	43.75%
SVM	44.30%	40.42%
KNN	40.51%	38.12%
Decision Tree	36.71%	35.74%

Observation: the Excellent class had only 19 students, creating a strong class imbalance and poor recognition of that class.

IMPROVED 3-CLASS MODEL PERFORMANCE

After grouping performance into Low / Average / Good, performance improved significantly.

Model	Accuracy	F1-Score
Decision Tree	87.34%	87.31%
Logistic Regression	86.08%	86.19%
Random Forest	86.08%	86.16%
Gradient Boosting	84.81%	84.89%
SVM	78.48%	78.36%
KNN	70.89%	71.07%

Key Finding: the three-class formulation significantly improved classification performance over the original four-class approach.

5-FOLD CROSS-VALIDATION

Model	Mean F1	Std F1
Random Forest	0.8534	0.0653
Gradient Boosting	0.8509	0.0297
Decision Tree	0.8342	0.0237
Logistic Regression	0.8312	0.0486
SVM	0.7979	0.0306
KNN	0.6637	0.0394

- Random Forest achieved the highest initial mean cross-validation F1-score.
 - Gradient Boosting showed very consistent fold performance.
 - KNN produced the lowest performance.
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HYPERPARAMETER TUNING

Tuned Random Forest

class_weight = balanced
max_depth = None
min_samples_split = 5
n_estimators = 300

Best CV F1-Score: 0.8700

Tuned Gradient Boosting

learning_rate = 0.1
max_depth = 2
min_samples_split = 10
n_estimators = 50

Best CV F1-Score: 0.8891

Tuned Decision Tree

class_weight = balanced
max_depth = 5
min_samples_leaf = 8
min_samples_split = 2

Best CV F1-Score: 0.8803

FINAL MODEL SELECTED: TUNED GRADIENT BOOSTING

Best Cross-Validation F1-Score: 0.8891

Metric	Result
Accuracy	87.34%
Precision	87.95%
Recall	87.34%
F1-Score	87.42%

Model	Accuracy	F1-Score
Tuned Gradient Boosting	87.34%	87.42%
Tuned Random Forest	87.34%	87.30%
Tuned Decision Tree	86.08%	86.02%

Conclusion: Tuned Gradient Boosting was selected as the final model because it achieved the highest test-set F1-score and the highest cross-validation F1-score among the tuned models.

TOP PREDICTIVE FEATURES — TUNED GRADIENT BOOSTING

Rank	Feature	Importance
1	G2 (2nd period grade)	90.78%
2	G1 (1st period grade)	1.71%
3	Age	0.94%
4	Going Out Frequency	0.78%
5	Weekend Alcohol Consumption	0.66%
6	Workday Alcohol Consumption	0.66%
7	Father's Job – Other	0.60%
8	Absences	0.60%
9	Past Failures	0.49%
10	Family Relationship	0.30%

Key Observation: G2 is the dominant predictive feature, followed by G1 — indicating previous academic performance is highly associated with the predicted category.

LIVE STUDENT PERFORMANCE PREDICTION

Example Student Input

- Age: 20
- Study Time: 1
- Past Failures: 1
- Absences: 25
- G1: 13 G2: 14
- Internet: Yes Higher Education: Yes
- School Support: Yes Family Support: Yes

Predicted Performance: GOOD

Performance	Probability
Average	39.29%
Good	53.53%
Low	7.17%

Interpretation: the model predicts that the student is most likely to belong to the Good performance category.

LIST OF MODULES (1/2)

Module 1 — User Authentication & Role Management

- Login, registration and JWT authentication.
- Admin / Faculty / Student roles.

Module 2 — Student Management

- Add, edit, delete, search and view student information.

Module 3 — Dataset Management & Preprocessing

- CSV upload, cleaning, missing values, duplicates, encoding and scaling.

Module 4 — ML Model Training & Comparison

- Train multiple models, evaluate metrics, compare models and select the best model.

Module 5 — Student Performance Prediction

- Predict High / Medium / Low academic performance for students.

Module 6 — Dropout Prediction

- Predict Low / Medium / High dropout risk using the approved input features.

LIST OF MODULES (2/2)

Module 7 — Risk Score & At-Risk Student Identification

- Generate risk scores and identify students requiring attention.

Module 8 — Analytics Dashboard

- Charts, statistics, performance trends and dropout-risk distribution.

IMPLEMENTED ADVANCED FEATURES — 1 to 3

1. Early Deterioration Detection — falling attendance, GPA and assignment/quiz performance.
2. Hidden / Emerging Risk Detection — identifies normal-looking students with declining engagement/behaviour.
3. Risk Trajectory Analysis — Improving / Stable / Increasing / Critical.

IMPLEMENTED ADVANCED FEATURES — 4 to 6

4. Explainable AI (XAI) — SHAP and feature importance explain why a student is at risk.
5. Personalized Intervention Engine — generates interventions from actual risk factors.
6. Reports & Administration — PDF/Excel/CSV reports, user/model management, audit logs and settings.

IMPLEMENTATION STATUS

All listed standard modules and the six advanced features above have been successfully implemented in the system.

LITERATURE SURVEY

Author(s) / Year	Title	Methodology	Inference
Nti & Ramanayake, 2026	Explainable ML for student dropout prediction and tailored interventions in online personalized education	Random Forest, XGBoost + SHAP	XGBoost reached 80% accuracy (recall 81.6%); explainability supports personalized intervention.
Martínez & Castillo, 2024	Prediction of student dropout using AI algorithms	Random Forest, Gradient Boosted Tree, Neural Network	Gradient Boosted Tree hit 95.44% accuracy using only the first 3 semesters.
Brand C. et al., 2024	Toward Educational Sustainability: An AI System for Identifying and Preventing Student Dropout	K-Means, Isolation Forest, Linear Regression, Decision Tree	Decision Tree reached 91% accuracy; small 288-student sample limits generalization.
Ovtšarenko, 2026	Using AI to Forecast Student Dropout Risk in Technical Education via Learning Analytics	Logistic Regression, Random Forest	Random Forest achieved 87.56% accuracy (F1 82.67%) from Moodle activity logs.
Wang, 2025	AI in student management systems for performance monitoring and intervention	Decision Tree, RF, SVM, ANN; hybrid voting/stacking	Hybrid models reached 96–99% accuracy across three public datasets.
Khatun et al., 2025	Student dropout prediction using machine learning and explainable AI	Random Forest, XGBoost, Logistic Regression; SHAP/LIME	XGBoost reached 94.41% accuracy (F1 96.62%) on a 71,531-sample national dataset.
Mohanasundharam et al., 2026	AI-Based Student Dropout Prediction and Counselling System	Logistic Regression, Decision Tree, Random Forest	Random Forest reached 95% accuracy; preprint, dataset size unclear, no XAI yet.
AlFaress, Janarthanan & Dixit, 2025	Impact of Data Feature Selection on AI-Based Student Performance Prediction	DT, SVM, NN, LR, RF, DKT, SAKT, Transformer	Feature selection strongly affects accuracy, fairness, privacy and interpretability.
Embarak & Hawarna, 2024	Enhancing Student Success with XAI-Powered RADAR for Early Detection of At-risk Students	Decision Tree with CFS feature selection; XAI	Reached 82.22% accuracy (F1 83.33%); generalization limited to one computing-major dataset.

LITERATURE SURVEY (CONTINUED)

Author(s) / Year	Title	Methodology	Inference
Ahmed, 2024	Student Performance Prediction Using Machine Learning Algorithms	K-Means clustering for feature analysis; SVM, Decision Tree, Naïve Bayes, KNN	SVM reached 96% accuracy after parameter tuning; Decision Tree second at 93.4%.
Guevara-Reyes et al., 2025	Machine learning models for academic performance prediction: interpretability and application in educational decision-making	XGBoost, Random Forest with hyperparameter optimization	Achieved $R^2 > 0.90$ and F1-score > 0.85 ; emphasizes interpretability for institutional decisions.
Wang & Yu, 2025	Machine learning approach to student performance prediction of online learning	Logistic Regression with Taylor-expansion optimization on correlation-filtered behavioural indicators	Outperformed four baseline classifiers; learning duration significantly affects predicted performance.
Bouallegue, Omri & Al-Naemi, 2026	Machine Learning Approaches for Early Student Performance Prediction in Programming Education	Six ML models incl. XGBoost; cross-validation with hyperparameter tuning	XGBoost exceeded 91% accuracy; midterm grade and lab completion time were top predictors.
Roy, 2026	Early Prediction of Student Academic Performance Using Machine Learning and Clustering Techniques	Preprocessing, feature selection, clustering-based segmentation, hyperparameter optimization	Proposes a scalable framework validated with Accuracy, Precision, Recall, F1 and ROC-AUC.
Zhu, 2024	Prediction and Analysis of University Student Performance Based on Machine Learning	Random Forest regression with feature-importance analysis (395-student dataset)	$R^2 = 0.55$, RMSE = 4.68; 2nd unit-test score was the strongest predictor of final performance.

RESEARCH GAP

RESEARCH GAP

- ☐ • Many studies focus on performance prediction or dropout prediction independently.
 - ☐ • Early deterioration, hidden/emerging risk and risk trajectory analysis are less commonly integrated.
 - ☐ • Explainability is often limited or absent.
 - ☐ • Prediction systems do not always connect risk detection with personalized interventions.
 - ☐ • End-to-end risk monitoring, reporting and administration remain limited.
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- ☐ **Our Contribution**
 - ☐ • Integrates performance prediction and dropout-risk prediction.
 - ☐ • Compares multiple ML models and selects the best-performing model.
 - ☐ • Implements early deterioration, hidden/emerging risk and risk trajectory analysis.
 - ☐ • Uses SHAP-based Explainable AI to explain risk factors.
 - ☐ • Provides personalized interventions, reports and administration features.
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CONCLUSION

An integrated AI-based Student Performance & Dropout Risk Prediction System was developed.

- The system includes student management, dataset preprocessing, ML model training and comparison, performance prediction and dropout-risk prediction.

- Successfully implemented advanced capabilities:

- • Early Deterioration Detection

Note: the current phase predicts student performance (Low/Average/Good) as the early identification module; dropout-risk prediction and intervention are planned as future work using dropout-labelled data.

- • Hidden / Emerging Risk Detection

Thank You